

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Master of Computer Applications (MCA)
Semester-II University Examination May 2010
CA706 Object Oriented Concepts and Programming in Java (JSE)

Date: 18.05.2010, Tuesday Time: 12:30 p.m to 3:30 p.m Maximum Marks: 70

Instructions:

1. Attempt the two sections in separate answer books.
2. Figure to the right indicate marks.
3. Make suitable assumptions wherever necessary and state them.

SECTION-1

- Q1 Do as directed:**
- (A) Fill in the blanks:** [5]
- 1 _____ operator is used to create an array.
 - 2 _____ method can be called without the instance of a class.
 - 3 When an interface method is implemented in a class, it must be declared as _____.
 - 4 _____ method of Math class returns the largest integer number less than or equal to given double value.
 - 5 Strings that need modification are handled by _____ class.
- (B) Differentiate the following:** [6]
- (i) Type Casting & Type Conversion
 - (ii) Applet & Application
 - (iii) 'this' & 'super'
- (C) What is a difference between Method Overloading & Method Overriding? Explain it with proper example.** [4]
- Q2 Do as directed:**
- (A) What are Encapsulation, Inheritance and Polymorphism?** [3]
- (B) What is wrapper class? Explain the Character class with any two methods with example.** [3]
- (C) How logical AND (&) and short-circuit AND (&&) are differ from each other?** [2]
- (D) Answer the following questions:** [2]
- (i) What is the use of getRuntime() ?
 - (ii) What is a Native method?
- OR**
- Q2 Do as directed:**
- (A) Explain the terms bytecode and JVM.** [3]
- (B) Explain any two methods by which string can be modified.** [3]
- (C) What is Instance variable hiding?** [2]
- (D) Answer the following questions:** [2]
- (i) What is the use of new () keyword?
 - (ii) Explain various types of comments supported by Java.
- Q3 Do as directed:**
- (A) Explain various access modifiers and their visibility.** [4]
- (B) Explain final class, abstract class and super class.** [3]
- (C) Give difference between String & StringBuffer.** [2]
- (D) Casting between primitive types allows conversion of one primitive type to another. (True / False)** [1]

OR

- Q3 Do as directed:**
- (A) Explain Interface with syntax & example. Also differentiate abstract class & interface. [4]
 - (B) Write a Java program that demonstrates multiple inheritance. [3]
 - (C) What are different types of inner classes? [2]
 - (D) What is the difference between assignment & initialization? [1]

SECTION-2

- Q4 Do as directed:**
- (A) **Answer the following questions:** [6]
- 1 The File class can be used to read & write data. (true/ false)
 - 2 What method is used to specify a container's layout?
 - 3 Name two subclasses of the TextComponent class.
 - 4 Write the output from the following block of code:
import static java.lang.System.*;
class demp
{
 public static void main(String a[])
 {
 out.println(12);
 }
}
 - 5 What is the purpose of the System class?
 - 6 Which class is the immediate superclass of the Container class?
- (B) **Differentiate the following:** [4]
- (i) Thread & Process
 - (ii) Checked exception & Unchecked exceptions
- (C) What is a Layout Manager? Explain the different Layout Managers. [5]
- Q5 Do as directed:**
- (A) What is the use of Synchronization? By which way we can achieve the functionality of it? [4]
 - (B) Write a Java program that copies one text file to another, converting every lower case character to upper case equivalent. Names of sources file and destination file should be passed from command line. [5]
 - (C) What is the Set interface? [1]

OR

- Q5 Do as directed:**
- (A) How the Inter Thread Communication can be done? Explain with example. [4]
 - (B) Write an Applet that will take the two Integer numbers from the user as a input and will display the greater number from it. [5]
 - (C) Which attributes are mandatory when creating HTML to display an applet? [1]
- Q6 Do as directed:**
- (A) What is the Vector class? Difference between Vector and ArrayList? [2]
 - (B) Explain different types of events and its listeners. [3]
 - (C) What is Exception Handling? Explain the exception handling with try, catch & finally block with the example. [5]

OR

- Q6 Do as directed:**
- (A) What is a Stream? Differentiate between Byte Oriented & Character [2]

Oriented Stream.

- (B) Can applets communicate with each other? How? Explain with example. [3]
- (C) Write a Java program that takes filename or directory name from command line. If it is a valid filename then it should display its size. If it is a directory then it should display the list of files of that directory. [5]